

NovaDrive Phase 1: network mapping and risk assessment

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Phase 1 mapped 24 suppliers across three tiers behind all three products, found four hidden choke points, and ranked every supplier on six risk dimensions.

The 1-minute version

- **The chain:** NovaDrive buys parts for 3 products (P1 DriveCore, P2 ChargeBridge, P3 StoreLink). Tier-1 suppliers sell directly to NovaDrive. Tier-2 suppliers sell to a Tier-1. Tier-3 suppliers sell to a Tier-2.
- **What we mapped:** 24 suppliers (8 Tier-1, 10 Tier-2, 6 Tier-3). They come from 32 supplier links in the data: 31 confirmed by a direct statement and 1 inferred (Verdant to IonPeak).
- **Main hidden problems:** one die source (IonPeak) sits behind both power-assembly suppliers; those two suppliers have the same parent company; one company (Jade) is the only qualified circuit-board substrate source; five network sites sit in the same flood-watch zone (Z01).
- **Money at stake:** all 3 products depend on IonPeak, which puts USD 1,560m of yearly product revenue behind one site.
- **Risk scoring:** every supplier gets six scores (0 to 100, where 100 is riskiest) that are combined into one Overall Risk %.
- **Top confirmed risks:** Jade (68.3%), IonPeak (64.7%), Orion (58.7%) and Meridian (57.8%).
- **High score, low confidence:** Verdant and Umber also rank high, but mostly because data is missing. They must be verified first.

Files that go with this doc

- **NovaDrive_Supplier_Tiering.xlsx** : every link with its evidence quote, tiers by product, shared dependencies, zone concentration, rejected links, unresolved items.
- **NovaDrive_Supplier_Risk_Scorecard_v2.xlsx** : the risk table, the supply paths, the scales and weights (editable), events, and a sensitivity test.

Part 1: How we mapped the supply network

The goal was to see who really sits behind NovaDrive's parts, down to Tier-3, using only the data we were given. We did it in six steps.

1. **Used only the supplied sheets.** Supplier Universe, Relationship Evidence and Components Context for the network. Business Context only for product revenue. No outside information.
2. **Read all 59 evidence rows one by one** and gave each link one label. *Confirmed*: a direct statement says A sells to B for a named component. *Inferred*: the record points that way but does not name the line. *Hypothesis*: only a lead (a sample, a directory listing), so it is not placed in the network.
3. **Matched by ID, never by name.** Many names look alike (Ion Peak Trading vs IonPeak Semiconductor, Delta Consumer Plastics vs Delta Capacitor Works). We matched on the ORG ID and the Site ID, and checked the component, date and product.
4. **Built the chains from NovaDrive downward.** Tier-1 comes from the supply-award records, which give each supplier's planning share of a component. Then we added who sells to each Tier-1, then who sells to them. Tiers are counted from NovaDrive, per component.
5. **Rejected what does not count.** Nine items were set aside: an expired housing award, a trial that never became supply, a broker and a research site with look-alike names, a freight forwarder, a port operator, a holding company, a head-office record, and the idea that two power-assembly suppliers mean two die sources. Each rejection has its reason and quote in the tiering file.
6. **Checked our own work.** Every quote was tested as a word-for-word copy of the evidence, every ID was confirmed to exist, and no expired record was used. All checks passed.

Which Tier-1 supplier makes which component

Component	Used in	Tier-1 supplier (planning share)	Time to qualify a new supplier
M10 Standard power assembly	P1, P3	Aster 60%, Boreal 40%	16 weeks
M20 High-voltage power assembly	P2	Aster 70%, Boreal 30%	16 weeks
C10 Control board	P1, P2, P3	Cobalt 100%	12 weeks
C20 DC-link capacitor	P1, P2	Delta Capacitor Works 100%	8 weeks
H10 Protective housing	P1	ForgeLine 100%	4 weeks
H20 Cooling assembly	P2, P3	Estuary 100%	6 weeks

Component	Used in	Tier-1 supplier (planning share)	Time to qualify a new supplier
B10 Battery-monitoring board	P3	Grove 80%, HarborSense 20%	12 weeks

Who sits behind them (Tier-2 and Tier-3)

Tier	Supplier	Sells to	Status
Tier-2	IonPeak Semiconductor	Aster, Boreal (power dies)	Confirmed
Tier-2	Lumen Magnetics	Aster, Boreal	Confirmed
Tier-2	Orion Ceramics	Aster, Boreal	Confirmed
Tier-2	Jade Printed Circuits	Cobalt, Grove, HarborSense	Confirmed
Tier-2	Kestrel Microdevices	Cobalt, Grove	Confirmed
Tier-2	Rill Connectors	Cobalt	Confirmed
Tier-2	Meridian Dielectrics	Delta Capacitor Works	Confirmed
Tier-2	Nacre Foils	Delta Capacitor Works	Confirmed
Tier-2	Pine Thermal Metals	Estuary	Confirmed
Tier-2	Quartz Alloy	ForgeLine	Confirmed
Tier-3	Umbra Silicon Carbide	IonPeak	Confirmed
Tier-3	Verdant Process Gases	IonPeak	Inferred
Tier-3	Yarrow Specialty Minerals	Orion	Confirmed
Tier-3	Warden Copper Foil	Jade	Confirmed
Tier-3	Xenon Resin	Jade	Confirmed
Tier-3	Zenith Polymer	Meridian	Confirmed

The chains stop at Tier-3 because the data says nothing about who supplies the Tier-3 companies. We did not guess.

Part 2: The pain points we found

On paper NovaDrive looks well spread out: two suppliers for power assemblies, two for battery boards. Underneath, many chains meet at the same few places. If one of these fails, several Tier-1 suppliers fail together.

Priority	Where	Why it hurts	Evidence
Critical	IonPeak Semiconductor (Tier-2, zone Z01)	The only qualified source of power dies behind both Aster and Boreal. No second die source is qualified. All 3 products depend on it (USD 1,560m yearly revenue).	DOC-078
Critical	CommonSpan Holdings (parent of Aster and Boreal)	The two "competing" power-assembly suppliers have the same owner and the same three Tier-2 suppliers (IonPeak, Lumen, Orion). Two Tier-1s, one real source.	DOC-075
High	Jade Printed Circuits (Tier-2, zone Z01)	The only qualified circuit-board substrate source for Cobalt's control board (all 3 products) and for both battery-board suppliers (P3).	DOC-079
High	Zone Z01 (East Delta)	Five network sites sit in the same zone: IonPeak, Jade, Orion, Umber and Verdant (inferred). One flood or outage can hit several chains at once. A flood watch (EV-001) is already active.	DOC-077, EV-001
Medium	Kestrel Microdevices (Tier-2, zone Z05)	Supplies microcontrollers to both Cobalt (control board) and Grove (battery board), so P3 depends on it twice.	DOC-031, DOC-033
Medium	Single-supplier components	C10, C20, H10 and H20 each have one Tier-1 supplier at about 100% of requirement. No backup is disclosed.	DOC-009, 011, 013, 015

The percentages are planning allocations stated in the records, not purchase history.

Part 3: How we scored the risk

Each of the 24 suppliers gets one Overall Risk %. A higher number means a riskier supplier. We built it in four steps.

1. **Pick the facts** that matter in six areas (the table below). All facts come from the supplied workbook.
2. **Put every fact on the same 0 to 100 scale** (100 = riskiest). We stretch each fact between the lowest and highest value found across all 30 companies in the indicators sheet. Where a high value is bad (debt), the highest value gets 100. Where a low value is bad (current ratio, on-time %), the lowest value gets 100. Example: Meridian has the lowest current ratio in the data (0.62), so it scores 100 on that fact.
3. **Average the facts inside each area** to get six scores, D1 to D6.
4. **Combine the six scores with weights** that add up to 100%.

$$\text{Overall Risk \%} = \frac{20\text{\,D}_1 + 15\text{\,D}_2 + 30\text{\,D}_3 + 15\text{\,D}_4 + 10\text{\,D}_5 + 10\text{\,D}_6}{100}$$

The six areas, how each is scored, and why it has its weight

Area	What it asks	How we scored it (0 to 100)	Weight	Why this weight
D1 Financial and operational health	Can this supplier keep delivering?	Average of: current ratio (low is bad), net debt / EBITDA (high is bad), on-time shipping % (low is bad), and the change in on-time % since January (falling is bad). Tier-3 suppliers have no financials, so only the two shipping facts are used.	20%	Basic health: a failing supplier is a risk wherever it sits.

Area	What it asks	How we scored it (0 to 100)	Weight	Why this weight
D2 Location and systemic exposure	Is its location exposed to shocks?	Average of the physical hazard, logistics friction and infrastructure indices of its zone. These are zone figures, so every company in a zone gets the same D2.	15%	Outside shocks matter, but they are less certain than an internal failure.
D3 Network dependence	Is it a hidden choke point?	Average of three facts: (a) how many Tier-1 suppliers an outage reaches (1 = 0, 3 = 100); (b) the share of the component's supply that flows through it; (c) single-source level: 100 if the evidence says no second qualified source exists, 50 if the data is silent or only one parent owns both, 0 if a second Tier-1 is disclosed.	30%	The case is about dependencies leaders cannot see, so this carries the most weight.

Area	What it asks	How we scored it (0 to 100)	Weight	Why this weight
D4 Business criticality and substitutability	How much revenue is tied to it, and how hard is it to replace?	Average of: (a) revenue of the products it reaches divided by the USD 1,560m total (P1 624, P2 520, P3 416); (b) the supplier qualification lead time divided by the longest one (16 weeks).	15%	A risk should be weighed by its impact on NovaDrive.
D5 Dynamic external threats	Is a live event hitting it?	The highest severity of a matching event: flood watch 40, financing pressure 60 (a confirmed shutdown would be 100). No match = 0.	10%	Only 8 events exist and just 2 match a supplier.
D6 Information gap and assurance	What do we not know?	Average of: the assurance gap index (100 if missing), the share of 8 indicator fields that are missing, and a link-evidence penalty (100 if the link is only inferred, 50 if one source family backs it, 0 if two or more).	10%	The handbook says missing information must never be treated as low risk.

How events were handled. Only events that really point at a supplier or zone count.

Event	Decision	Reason
EV-001 East Delta flood watch	Scored 40 for every Z01 supplier	A watch, no damage confirmed
EV-003 Meridian financing pressure	Scored 60 for Meridian	Late payments and covenant talks, production continues
EV-002 repeat of the flood watch	Ignored	Same bulletin, no new information
EV-004 Ion Peak Trading insolvency	Ignored	A broker, not IonPeak Semiconductor
EV-005 old port strike	Ignored	A 2023 event that ended
EV-007 Jade head office moved	Ignored	Production site did not change
EV-009 Delta packaging allegation	Ignored	About Delta Consumer Plastics, not Delta Capacitor Works
EV-010 Gulf Arc port delay	Not scored for now	No supplier or shipment in the data is tied to that port, and the impact is not assessed

How tie-ups affect the score. Risk travels up the chain, so we did not copy a Tier-3 score onto its Tier-2 and Tier-1 buyers. Instead D3 and D4 give high scores to any node whose failure hits many Tier-1 suppliers, many products and long replacement times. That is why IonPeak and Jade rank near the top even though their own finances are only middling.

Confidence is shown separately from risk. A supplier's confidence is the lower of two things: how well its link is backed (High with two or more independent source families, Medium with one, Low if only inferred) and how complete its data is (High only if no indicator field is missing). The risk band is relative: the top 8 are High, the next 8 Medium and the last 8 Low.

Part 4: The risk table

Sorted by Overall Risk %. D1 to D6 are the six scores (0 to 100). Risk band: ranks 1 to 8 High, 9 to 16 Medium, 17 to 24 Low. The full driver text for every supplier is in the scorecard file.

Rank	Supplier (ID)	Tier	D1	D2	D3	D4	D5	D6	Overall %	Confidence	Main driver
1	Jade Printed Circuits (ORG-453)	T2	41	76	100	88	40	15	68.3	High	Only qualified substrate source (DOC-079); outage hits 3 Tier-1s; zone Z01
2	Verdant Process Gases (ORG-736)	T3	41	76	67	100	40	79	66.6	Low	Link only inferred; data missing; zone Z01
3	Umber Silicon Carbide (ORG-455)	T3	49	76	67	100	40	62	66.5	Low	Data missing; one source; zone Z01
4	IonPeak Semiconductor (ORG-439)	T2	40	76	83	100	40	12	64.7	High	Only die source (DOC-078); hits both power Tier-1s; zone Z01
5	Orion Ceramics (ORG-708)	T2	28	76	67	100	40	27	58.7	Medium	Zone Z01; reaches all 3 products; 16-week replacement
6	Meridian Dielectrics (ORG-210)	T2	100	23	50	62	60	41	57.8	Medium	Weakest finances (current ratio 0.62, debt/EBITDA 7.2, on-time down 17.6 pts); EV-003
7	Warden Copper Foil (ORG-873)	T3	17	60	83	88	0	63	56.7	Low	Feeds 3 Tier-1s through Jade; no financials in data
8	Xenon Resin (ORG-179)	T3	17	57	83	88	0	63	56.3	Low	Feeds 3 Tier-1s through Jade; no financials in data
9	Lumen Magnetics (ORG-942)	T2	31	57	67	100	0	38	53.6	Medium	All 3 products; 16-week replacement; hits both power Tier-1s
10	Yarrow Specialty Minerals (ORG-520)	T3	15	42	67	100	0	62	50.6	Low	Data missing; all 3 products; 16-week replacement
11	Cobalt Control Electronics (ORG-119)	T1	35	57	67	88	0	14	50.0	High	Only C10 supplier (100%); 12-week replacement
12	Kestrel Microdevices (ORG-728)	T2	21	24	67	88	0	35	44.5	Medium	All 3 products; hits 2 Tier-1s
13	Nacre Foils (ORG-849)	T2	44	60	50	62	0	24	44.3	Medium	Current ratio 1.54; on-time 79.3% and falling; zone Z06
14	Rill Connectors (ORG-469)	T2	19	57	50	88	0	36	44.0	Medium	All 3 products; 12-week replacement

Rank	Supplier (ID)	Tier	D1	D2	D3	D4	D5	D6	Overall %	Confidence	Main driver
15	Aster Power Assemblies (ORG-247)	T1	28	27	57	100	0	8	42.4	High	16-week replacement; same parent as Boreal
16	Boreal Power Systems (ORG-725)	T1	36	24	47	100	0	11	40.9	High	16-week replacement; same parent as Aster
17	Pine Thermal Metals (ORG-274)	T2	33	60	50	49	0	30	40.9	Medium	Zone Z06; debt/EBITDA 3.67
18	Estuary Thermal Systems (ORG-786)	T1	11	60	67	49	0	20	40.6	High	Zone Z06; only H20 supplier
19	Delta Capacitor Works (ORG-922)	T1	21	23	67	62	0	17	38.6	High	Only C20 supplier (100%)
20	Zenith Polymer (ORG-659)	T3	12	23	50	62	0	62	36.3	Low	Data missing; one source
21	ForgeLine Enclosures (ORG-287)	T1	25	27	67	32	0	23	36.3	High	Only H10 supplier (100%)
22	Grove Battery Controls (ORG-103)	T1	40	57	27	51	0	6	32.9	High	On-time 78.8%; zone Z04
23	Quartz Alloy (ORG-329)	T2	20	27	50	32	0	33	31.2	Medium	Nothing stands out
24	HarborSense Electronics (ORG-176)	T1	30	24	7	51	0	9	20.1	High	Nothing stands out

How to read it

- **Jade and IonPeak are the top risks we can rely on.** Both have direct evidence of no second source, High confidence, and sit in the flood-watch zone.
- **Meridian is the only financially stressed supplier**, with the one active financing event.
- **Verdant, Umber, Warden, Xenon and Yarrow rank high partly because data is missing.** They are Low confidence, so verify them before acting.
- **Tier-1 suppliers rank lower** because they are well documented and mostly healthy. Their risk is mainly being the only supplier for a component.

What to trust, what we assumed, and what changed

Safe to rely on. The 31 confirmed links, each backed by a word-for-word quote. The single-source facts for IonPeak (DOC-078) and Jade (DOC-079). The Aster and Boreal shared parent (DOC-075). The 0 to 100 scaling, which is plain arithmetic on the supplied indicators.

Our own judgement, not data. The data gives no weights, so these are choices we made and can defend:

- The six weights (20, 15, 30, 15, 10, 10).
- The event severities: flood watch 40, financing pressure 60, confirmed shutdown 100.
- The single-source levels: 100, 50, 0.
- Scoring a missing assurance gap as 100, so missing data never looks safe.
- The High, Medium and Low bands, which are relative thirds of the 24 suppliers.

These are all editable on the Scales sheet of the scorecard file. We also tested equal weights and a health-led set. The top confirmed risks stay near the top, but some mid-table suppliers move several places, so treat close scores as ties.

Limits of the data.

- Allocation percentages (60/40, 100%) are planning shares, not purchase history. There is no spend, price, quality or capacity data.
- Tier-3 suppliers have no financial figures.
- Zone indices are the same for every company in a zone.
- The chains stop at Tier-3. The evidence IDs are not continuous, so the sheet looks like a sample, and the case cut-off date is not stated.
- Two leads are held back as hypotheses and not scored: Solace Optics to Grove, and Alder Bauxite to Quartz.

What changed from our earlier notes. Please check these with the team.

Topic	Earlier notes	Final model	Why
Weights	20 / 15 / 25 / 20 / 10 / 10	20 / 15 / 30 / 15 / 10 / 10	We moved 5 points from business criticality to network dependence, because hidden shared dependencies are the core of the case. Part of criticality is already counted inside D3. Easy to switch back.
Missing data	Fill with the 65th percentile, or add a flat 15-point penalty	Missing assurance gap scored 100, plus the share of missing fields, plus a link-evidence penalty, all inside D6	Simpler to explain, and no invented numbers go into D1.

Topic	Earlier notes	Final model	Why
Flood watch EV-001	Spike the location or hazard score	Scored 40 in the separate events dimension (D5)	Avoids counting the same flood twice in D2 and D5.
Gulf Arc delay EV-010	Raise logistics risk for suppliers using that port	Not scored	No supplier or shipment in the data is linked to Gulf Arc. We will revisit it if one is found.
Risk inheritance	Pass a Tier-3 problem up to Tier-2 and Tier-1	Handled through D3 and D4	Upstream nodes that feed many Tier-1s and products score high on their own, with no double counting.

Next steps

1. **Find alternate suppliers from public web sources** for the four confirmed top risks: Jade (substrate), IonPeak (power dies), Orion (ceramics) and Meridian (dielectrics). Screen them with the "Key Criteria for Alternate Supplier Search" column in the Components Context sheet. Score fit separately from risk. PolarSwitch is named in DOC-078 only as a prospective die source and is not qualified.
2. **Verify the low-confidence names** (Verdant, Umber, Warden, Xenon, Yarrow) by looking for a second source or direct disclosure. Verdant's link to IonPeak is still only inferred.
3. **Build the pitch deck** from the network map, the pain points, the risk table and the alternate-supplier shortlist.